

TBAILEY LLC · SUBMERGED ARC WELDING

Automatic Height Control for SubArc Seam Welding

Holding contact-tip-to-work distance constant on long straight and circumferential seams — as booms sag and cans walk — to protect weld quality on thick tank and pressure-vessel work, using our very own slide controller.

Prepared for
Senior Management

Date
July 2026

HOLDS

Constant CTWD, pass over pass

IGNORES

Tacks, gaps & spatter — no chasing

BUILT ON

TBailey's patented method, in-house

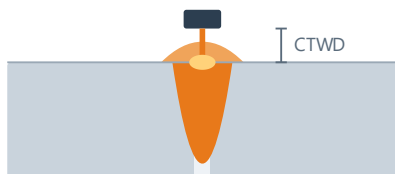
01 THE PROBLEM

CTWD drifts on every long seam

Our SAW process is unforgiving of contact-tip-to-work distance (CTWD): a small change in the torch's height changes the arc, the penetration, and the bead — and on thick, multipass tank and pressure-vessel plate that error compounds pass over pass.

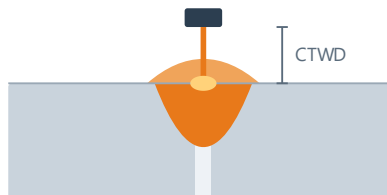
A LITTLE HEIGHT ERROR → A DIFFERENT WELD

CTWD TOO LOW



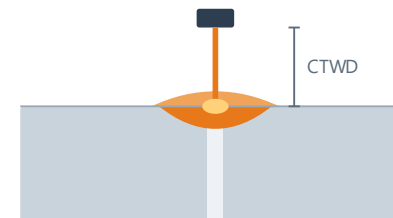
Runs hot — deep penetration
Excess melt-through, undercut risk.

ON TARGET



Sound fusion
Correct penetration and bead profile.

CTWD TOO HIGH



Runs cold — shallow bead
Lack of fusion at the root.

...AND ON OUR SEAMS THE HEIGHT NEVER HOLDS STILL

Booms sag as they travel

On long straight seams the boom cantilevers further out and deflects — the torch drops as the weld progresses.

Cans walk & sag on the rolls

On girth seams the can is never perfectly round; it walks and settles as it turns under the torch.

By hand it's a losing game

Holding CTWD manually across long, slow seams is fatiguing — small drifts are missed until the bead shows it.

A laser that watches, a controller that reacts

A Keyence laser measures CTWD continuously. Our **in-house slide controller** reads it and drives the welding machine's vertical slide up or down to hold the set height — correcting boom sag on straight seams and can walk/sag on circumferential seams alike. No new motor, no new mechanics.



Set it on the amplifier

Target CTWD and tolerance are dialed in on the Keyence amplifier — the shop's familiar control surface.



Works on both seam types

The same loop holds height as the boom extends on a straight seam or as a can turns under the torch on a girth seam.



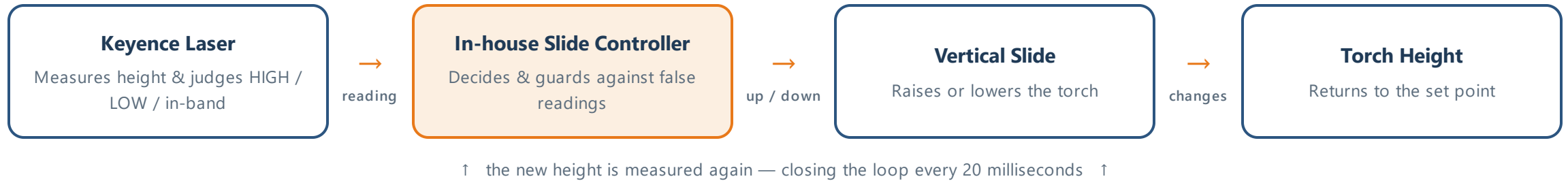
Isolated & safe

Drives the machine's own slide through electrically isolated interfaces; the welding circuit stays separated.

One operator button turns it **on and off**. One status light tells the whole story. Everything else is automatic.

A continuous measure-decide-move loop

Fifty times a second, the system measures CTWD, decides whether to raise, lower, or hold, and acts. Slow drift — a sagging boom or a settling can — is followed smoothly; the operator just sees the torch holding height across the seam.



The judgment does the fine control

The amplifier reports too-high / too-low / in-band. That starts a correction and picks the direction.

The live distance decides when to stop

The torch keeps moving until the measured distance is back at center — so it settles at the true target, not the edge of the band.

It knows the difference between the plate and an obstacle

Real plate-height changes are slow and smooth. Tacks, gaps, spatter, and running off the plate are sudden or invalid. The controller uses that difference to protect the torch.

- It constantly watches **how fast** the reading is changing and whether it is **valid**.
- A sudden jump or an invalid reading means "**this is not a real height change**" — the controller **freezes** and holds the torch exactly where it is.
- It **resumes only** when the reading is stable again and **back at the height it had before** the disturbance.
- Because the plate on both sides of a tack or gap is at the same height, resuming means it is **already correct** — no dive, no climb.

TRACKING — following the plate

Slow, valid readings. The torch gently follows genuine plate variation and keeps its **baseline** up to date.

HOLD — riding over a disturbance

A fast jump, an out-of-range reading, or a sensor alarm freezes motion. It coasts across at constant height and resumes on return to baseline.

What it handles — and how

Event	Response	What happens
Tack weld (raised bump)	Hold	Freezes and coasts over at constant height; resumes once past. It never climbs the tack.
Gap or gouge	Hold	Holds through the void and resumes on the far side, at the same plate height.
Laser runs off the plate / no target	Hold	Holds the last good height — never dives into the void; resumes when the plate is back under the beam.
Weld spatter / momentary flash	Filtered	Rejected before it can affect anything, by averaging many readings each cycle.
Genuine new plate height	Adapt	If the height really changes and <i>stays</i> changed — longer than any tack could last — the new height is accepted and tracked.

The "stays changed" window (~7 seconds) is deliberately set longer than the longest time a tack can sit under the beam at production travel speeds, so a tack is never mistaken for a real step.

Fails to a stop, never to a lurch

Starts OFF

On power-up the system is off. The operator must deliberately press to engage — it never wakes up moving.

Fails to a stop

If the sensor faults or loses the plate, motion stops and the torch holds. It does not guess.

Operator in control

The slide only responds while the operator has Auto Height switched on; off returns to normal manual control.

Safe on reset

After any power cycle or reset, all outputs come up off first, so the slide cannot lurch.

No conflicting commands

The controller can physically never call for "up" and "down" at the same time.

Electrically isolated

Every link between controller and welding machine is optically isolated — the welding circuit stays separated.

One button. One light. That's the interface.

There is nothing to configure at the machine during a shift. The operator presses to engage and reads a single status light.

The Auto Height button

Press once to turn **ON**, press again to turn **OFF**. On means the controller may move the slide; off means normal manual control.

What to expect

The torch gently follows the plate. Crossing a tack or gap, the light blinks for a moment while it holds steady, then goes back to tracking. No operator action needed.

The status light

-  **Off** — normal, tracking the plate.
-  **Fast blink** — holding on purpose, riding over a tack or gap. Resumes on its own.
-  **Solid on** — sensor alarm: the laser can't read the plate. Motion is stopped and held.

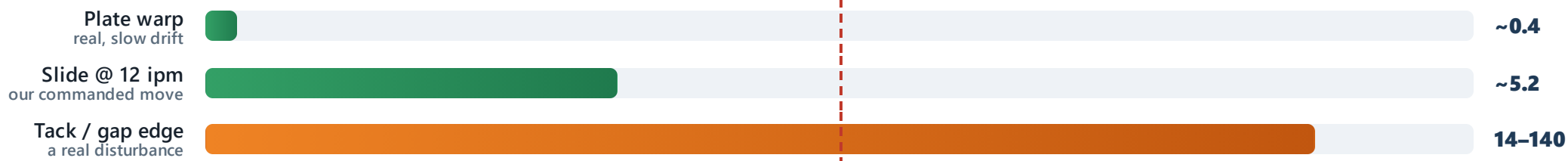
Target height and tolerance are set on the Keyence amplifier — the same panel the shop already uses.

Validated against our slide speed

The same guard that rejects fast disturbances must not mistake the slide's *own* commanded movement for one. So we plotted every source of reading-change against the freeze line — the slide has to fall *below* it, tacks and gaps *above* it.

Reading change **per 20 ms cycle** (sensor counts)

| dashed line = freeze trip (8 counts)



Green sits left of the line — followed smoothly; orange crosses it — caught and frozen. Our slide's own motion (5.2) clears the trip (8) with ~1.5× headroom, while tacks trip it every time.

How we checked it

12 in/min = 5.08 mm/s. At the amp's ± 10 mm window (≈ 51 counts/mm) and the 20 ms loop, that is ~ 0.10 mm — about **5.2 counts** — of change per cycle, versus the **8-count** trip.

Stays valid unless...

Documented to be re-checked if the slide ever runs faster than **~18 in/min**, or the laser's measuring window is rescaled to more counts/mm — either would raise the slide's per-cycle number.

Result: at 12 in/min the controller **will not fight itself** — a validated margin, not a hope.

Modest hardware, disciplined firmware

Proven industrial parts and a lean control program — built on a patented height-control method and maintained in-house, so the capability is ours to run, service, and tune.

HARDWARE



Keyence IL-300 / IL-1000

Proven industrial laser sensor and amplifier; does the height judging.



Compact controller

Runs the 50 Hz loop on a dedicated microcontroller — ~14% of program memory.



Machine's vertical slide

No new actuator — our controller drives the welder's own slide through isolated interfaces.

FIRMWARE



Patented control method

Based on TBAiley's patented method — a differentiated approach, not a generic off-the-shelf follower.



Observable, not a black box

Every input, output, and decision streams live during setup — verifiable and tunable.



Set once, then leave it

Sensitivity and timing fixed at commissioning; day-to-day only the amplifier is touched.

Proven sensor + a **patented control method** + lean, in-house firmware = a capability **TBAiley owns end-to-end.**

Status & path to the floor

The firmware is written, builds cleanly, and behaves correctly on the bench. The remaining work is structured commissioning on the machine.

✓ Complete

- Control loop & disturbance guard implemented
- Safety behaviors (start-off, fail-to-stop, interlock)
- Slide-speed margin analysis & documentation
- Clean build; live diagnostics in place

▷ Commissioning checklist

- Verify each signal's polarity & the fail-safe alarm
- Confirm raise/lower direction on the slide
- Tune disturbance sensitivity to real travel
- Low-stakes live weld trial, then production

Commissioning is done with the machine's slide disconnected first, then reconnected at low travel — no production risk during bring-up.

IN SUMMARY

Constant CTWD on every seam — hands-off, without being fooled

- Holds CTWD automatically through boom sag and can walk/sag
- Protects weld quality on thick, multipass tank and pressure-vessel seams
- Rides over tacks, gaps, and spatter instead of chasing them
- Works on both straight and circumferential seams
- Runs on our in-house slide controller; fails safe; operator in control
- Engineering-validated margins — one button, one light

Ask: Approve on-machine commissioning and begin training of operators.